MONTHLY PROGRESS REVIEW

Leveraging Hybrid RAG Pipelines for AI-Based Test Execution Insights in Test Management Systems

Worklet ID: 26SE08
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1. Project Objective & Proposed Solution

Hybrid retrieval for testing artifacts and execution insights



Vanilla RAG
- Query embedding
- Vector similarity search
- Top-K retrieval
- Relevance scoring

Graph RAG
- Entity extraction & linking
- Multi-hop graph traversal
- Relationship-aware retrieval
- Graph relevance scoring

Planner principle: Vanilla RAG runs first; Graph RAG is invoked only when Vanilla RAG relevance is below the threshold.

2. Last Month's Progress

Work completed after the initial project presentation

GitHub repository	**Completed**	Repository and project file structure established
System architecture	**Completed**	End-to-end Hybrid RAG flow defined
Vanilla RAG	**In progress**	Core pipeline prepared for prototype validation
Hybrid planner	**In progress**	Threshold-based routing logic defined
Database	**In progress**	Tentative storage requirements/schema identified
Samsung dataset	**Blocked**	Requested ; dataset still pending

3. KPI / Milestone Achievement

Status aligned with the initial worklet milestone

Requirements gathering & finalization	**IN PROGRESS**	Initial requirements captured through project discussions
Dataset collection & generation	**BLOCKED**	Samsung dataset pending
Architecture Design — UI / DB / API	**COMPLETED**	Architecture and module design prepared
System Flow Diagram	**COMPLETED**	Hybrid RAG system flow prepared
LLD / HLD documentation	**IN PROGRESS**	Architecture documentation prepared; implementation details being refined

4. Technical Challenge & Mitigation

Dataset dependency is the main blocker

CHALLENGE

Samsung testing dataset has not yet been received.

Requested; currently pending.

IMPACT

- **Real-data chunking validation**
- **Metadata/schema finalization**
- **Vector indexing with actual artifacts**
- **Knowledge graph construction**
- **Retrieval evaluation**

MITIGATION

Decouple dataset-dependent work from pipeline development by using a synthetic testing dataset for technical validation.

5. Current Technical Implementation — Vanilla RAG

Core semantic retrieval path being prepared for prototype validation

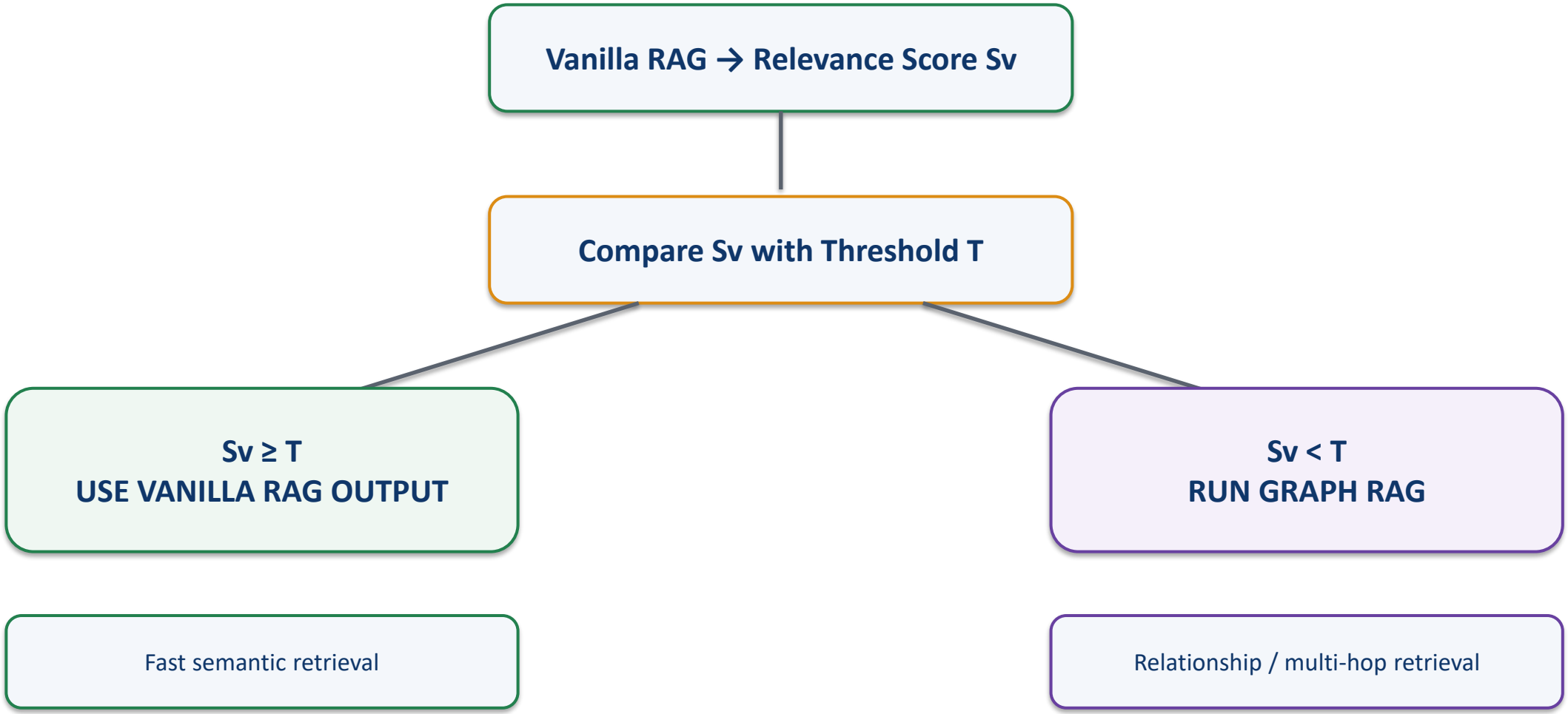


Example validation query
“What defects are related to automation?”

Synthetic test artifacts can be used to validate retrieval and scoring before the Samsung dataset arrives.

6. Hybrid RAG Planner — Threshold-Based Decision

Vanilla RAG first; Graph RAG only when semantic retrieval is insufficient



7. Results & Initial Observations

What we can establish before the production dataset arrives

RESULTS SO FAR

✓ GitHub repository established
✓ System architecture organized
✓ System flow documented
✓ Threshold-based planner logic defined
✓ Tentative database/storage design identified

OBSERVATIONS

- Vanilla RAG is the appropriate first retrieval layer for direct semantic queries.
- Schema should remain flexible until real dataset metadata is known.
- Synthetic data allows implementation to continue while the production dataset is pending.

These are initial implementation observations, not final performance results.

8. Upcoming Month's Plan

Move from architecture and prototype setup toward integrated implementation

01	**Dataset & DB**	Analyze Samsung dataset when received; finalize schema; implement chunk + metadata storage.
02	**Vanilla RAG**	Complete embeddings, vector search, Top-K retrieval and relevance scoring.
03	**Hybrid Planner**	Implement and validate threshold-based Vanilla → Graph routing.
04	**Graph RAG**	Implement entity extraction, graph construction, traversal and relationship-aware retrieval.
05	**Integration**	Connect selected context to response-generation module and begin end-to-end testing.
06	**Evaluation**	Define retrieval/relevance metrics and compare retrieval behaviour.

THANK YOU

Worklet 26SE08
Discussion & Feedback